

SKILL-4

4. Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    for(i=0; i<3; i++)
    {
        if(fork()==0)
        {
            printf("Child %d PID=%d\n", i+1, getpid());
            return 0;
        }
    }

    for(i=0; i<3; i++)
    {
        wait(NULL);
    }

    printf("All children completed.\n");

    return 0;
}
```

OUTPUT:

```
saiteja@LAPTOP-5BAATM70:~/TeacherFile-skill$ nano skill4.c
saiteja@LAPTOP-5BAATM70:~/TeacherFile-skill$ gcc skill4.c -o skill4
saiteja@LAPTOP-5BAATM70:~/TeacherFile-skill$ ./skill4
Child 1 PID=1042
Child 3 PID=1044
Child 2 PID=1043
All children completed.
saiteja@LAPTOP-5BAATM70:~/TeacherFile-skill$ |
```